Ryan Elisha Bantu

AI Systems Engineer | LLM Systems Engineer | Full-Stack Engineer

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Summary

AI and systems engineer specializing in embedded AI, LLM systems and full-stack product development. Built Edge AI pipelines (YOLO detection and Re-ID) for resource-constrained hardware and production LLM pipelines on AWS Bedrock reaching 99.8% output accuracy. Founder of GoPerch Innovations, serving 40+ enterprise clients. **Research interests:** Edge AI, Embedded ML Systems, Real-Time Inference, Multimodal Learning, Distributed Intelligent Systems, Wind Energy, Climate Change, Materials Science, Rare Earth Deposits, Semiconductors.

Education

Manipal Institute of Technology, Manipal **2021 – 2025**
B.Tech in Mechanical Engineering

FIITJEE Junior College – Intermediate Education (MPC) 2019 – 2021
Kendriya Vidyalaya – Secondary Education 2009 – 2019

Technical Skills

- **AI & Machine Learning:** Anthropic Claude, Llama 3.2, Qwen, LLM Fine-Tuning (SFT), Prompt Engineering, Evaluation Frameworks, YOLO (CNN), Object Detection, Re-Identification (Re-ID), Random Forests, EEG Processing
- **Cloud & Backend:** AWS Bedrock, AWS S3, FastAPI, Node.js, Fastify, REST APIs, PostgreSQL, MongoDB, SQLite, Firebase, Supabase, Neon, Render, CI/CD, Automated Testing
- **Languages & Frameworks:** Python, C, C++, C#, .NET 8, Godot 4, JavaScript, TypeScript, Swift (SwiftUI), Kotlin, Dart (Flutter), React.js, Next.js, MERN Stack, MATLAB
- **IoT & Embedded:** Raspberry Pi, Arduino, RFID Systems, SIM7600 (4G), Sensor Fusion, Real-Time Telemetry
- **Electronics & PCB Design:** KiCad, Altium, Multi-Layer PCB Design (Schematic Capture, Layout & Routing), Fabrication & Assembly
- **Mechanical & Simulation:** SolidWorks, Fusion 360, CATIA V5, HyperMesh, ANSYS Fluent (CFD), Altair OptiStruct

Research & Publications

- **IoT Framework for Real-Time Patient Monitoring in Emergency Transfers** *(Published)* – Presented at SCPGCON 2026, IEEE Bangalore Section, MIT Bengaluru (Sept 2026). Architected an edge-based telemetry system streaming patient vitals from ambulances with low-latency pipelines for pre-arrival clinical insight, evaluated under network and hardware constraints.
- **Day-Ahead Wind Power Forecasting for the Chalmers Björkö Turbine** *(In Process)* – Project page. Built a 15-minute, 24-hour-ahead forecast for a 45 kW research turbine with only 44 non-contiguous SCADA days: fitted an empirical PCHIP power curve from 20 Hz data and bias-corrected SMHI and Open-Meteo NWP wind via model output statistics (MOS). Achieved leave-one-day-out wind MAE of 1.003 m/s and power nRMSE of 0.274, cross-validated on the multi-year Kelmarsh wind-farm record.
- **EEG-Based Phoneme Extraction for Cerebral Palsy Communication** *(In Process)* – Trained deep learning models on EEG signals (Caravan dataset) to extract phoneme-level representations, with preprocessing and feature extraction robust to noise, enabling interpretable mappings for thought-to-speech synthesis.

Professional Experience

Windscapes Landscaping **03/2026 – Present**
Forward Deployed Engineer *Michigan, USA*

- Architecting an enterprise inventory management platform that improved operational efficiency by 64%, combining Flutter Admin and Employee apps, FastAPI and React services on AWS, and edge RFID modules (Raspberry Pi, Arduino) for real-time workforce tracking and automated payroll.

GoPerch Innovations Pvt Ltd **2021 – Present**
Founder & CEO *goperch.in*

- Lead technical architect and business head delivering end-to-end "Idea-to-Action" product development, scaling solutions from concept to deployment across **40+ enterprise clients**; manage and mentor a team of **8 engineers**.
- **Edge AI Wildlife Monitoring (ruggedlynx.com):** Architected autonomous trail cameras running YOLO for real-time animal detection and classification, with similarity analysis and Re-ID on Raspberry Pi and SIM7600 modules for remote data delivery to researchers worldwide.
- **ERP & CRM Systems:** Built custom ERP systems for multi-site landscaping operations and inventory, and a CRM for gaming centers that reached **42 installations in 6 months**.

- **Compliance & Smart Infrastructure:** Engineered inventory compliance platforms automating regulatory reporting for **15+ SMBs**; led a geo-fencing and geo-tagging parking reservation system for the Manipal campus.

Manipal Dot Net Pvt Ltd **11/2025 – 03/2026**
Intern – AI Systems *Manipal, India*

- Engineered AI clinical workflow platforms integrating Anthropic Claude on AWS Bedrock with fine-tuned open-weight LLMs (Llama 3.2, Qwen, II-Medical-8B) for automated checklist and diagnostic generation.
- Designed cost-aware selective regeneration pipelines with structured prompting (85% spec adherence), evaluation frameworks, scoring heuristics and retry/fallback logic, raising medical checklist accuracy to **99.8%**.
- Built FastAPI and MongoDB backend services with React.js frontends, and deployed AWS infrastructure for Bedrock APIs and production services.

Infopine **06/2025 – 08/2025**
Intern – PCB & ERP *India*

- Designed 4-layer PCBs, improved fabrication and assembly workflows, and built AI-powered ERP/CRM and inventory modules for textile clients (Spectrum, Linkage), contributing to **35% sales growth**.
- Onboarded clients in India and the USA (Zeeland and Holland, Michigan), generating **$350,000 in revenue** and delivering full-stack apps in Next.js, MongoDB, JavaScript and Python.

Sequoia Imports and Exports **08/2024 – 12/2024**
Full-Stack Engineer Intern *India*

- Built and deployed a MERN-stack business website with SendGrid email automation and LLaMA-powered agentic chatbots in Python for turmeric import/export.

DRDO (Defence Research and Development Organisation) **05/2024 – 08/2024**
Composite & Structural Engineering Intern, ASL / CPDC (AGNI, VEDA) *India*

- Designed and manufactured carbon-fiber composite components for missile and rocket systems, achieving 23% weight reduction, 17% better thermal resistance and 11% higher load capacity.
- Conducted structural validation testing that increased structural integrity by 37% and improved thermal shielding efficiency by 17% under extreme operating conditions.

Projects

Cardilyn – Family Health Companion cardilyn.com | GitHub

- Native iOS (SwiftUI) and Android (Kotlin) apps on Firebase with family workspaces, medication reminders, vitals logging, HealthKit/Health Connect sync, doctor–patient sharing and PDF reports. In TestFlight and Play closed testing.

Unearth – Map of Little-Known Public Places Website | GitHub

- iOS (SwiftUI), Android (Kotlin/Compose) and React clients on Fastify and PostgreSQL; no ratings, ads or ranking, as a place earns trust only when others independently find it. In TestFlight and Play closed testing.

farmXPdition – Productivity RPG Website

- Co-built a cross-platform 2D pixel-art RPG that turns real productivity into progress: projects become village doors, tasks become quests, and finishing real work earns XP and gold that grow a farming world.
- Built in C#/.NET 8 and Godot 4 on a local-first SQLite core with 280+ automated tests and Windows/macOS CI; shipped as a signed, notarized Mac release with in-app update notices and a React download site.

VAYU (CBEC) – Retrofit Cooling Device Website

- Award-winning fan-mounted cooler combining thermoelectric cooling and desiccant dehumidification, validated with CFD in ANSYS Fluent and Altair OptiStruct; sold 10 units with 20+ pre-orders.

SYPE – Assistive Communication Tool: React communication app for non-verbal autistic children; 4 daily users.
Titanic Survival Prediction: Random Forest built from Stanford CS229 (Andrew Ng) math on Kaggle data.

Achievements & Awards

- **National Winner, Solar Decathlon India 2024–25:** 1st place in the Residential Cooling Retrofit division with VAYU, outperforming 500+ participants from premier institutions.
- **Provisional Patent (No. 202541112613):** Filed for VAYU, a retrofit cooling device that reduced ambient temperature by 4 °C.
- **MAHE Start-Up Founder's Felicitation:** Honoured as Founder of GoPerch Innovations by the MAHE Innovation Centre and Institution's Innovation Council (news).
- **Forbes Marshall Award (Runner-Up):** Recognized for academic projects advancing energy and sustainability.

Leadership & Volunteering

- **Guest Lecturer, Infosys Bangalore:** Lectured on sustainability, innovation and team execution.
- **Math & Physics Instructor, BOC Society:** Prepared students for SAT, JEE and Olympiads; ran Python workshops for 50+.
- **Technical Leadership:** Organized TechTatva (30+ events); IEEE/ASME board member; raised INR 30,000+.